Extra Assignment – IT3011- Theory and Practices in Statistical Modelling

Duration: 5pm – 7pm on 11th June, 2021
Submission: Write answers in a separate paper, then scanned or take a photo and upload them on or before 11th June, 7.00pm

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a). Let X be a random variable. Define $f(x)=1/3,\ where\ 1\leq x \leq 4$

i. Prove that $f(x)$ is a probability density function.
ii. Find the mean and standard deviation of X.

b) Kevin is required to sell candy bars to raise money for the 8th grade field trip. There is a 45% chance of him selling a candy bar at each house. He has to sell 6 candy bars in all. Let X be of number of houses he visits.

i. Name the probability distribution (with parameter(s)) of X.
ii. What is the probability he sells his last candy bar at the 10th house?
iii. What is the probability that he finishes on or before the 9th house?
iv. What is the probability that he sells his 1st candy bar at the 5th house?

c) Let X be an amount of time (in minutes) a telephone operator spends with a customer. The time is known to have an exponential distribution with the average time five minutes.

i. Find the probability that a telephone operator spends four to five minutes with a randomly selected customer.
ii. Find the median of X
iii. Which is larger, the mean or the median?

d) It was found that the mean length of 100 steel bars produced by a company was 120.05 cm with a variance of 2.4 cm. Find the probability that a steel bar selected at random would have a length, [Write the R code along the answer]. Assume that the lengths of steel bars produced by a company follows a normal distribution.

a) Between 126.09 cm and 116.80 cm.
b) Less than 118.01 cm.
c) Greater than 125.49 cm.